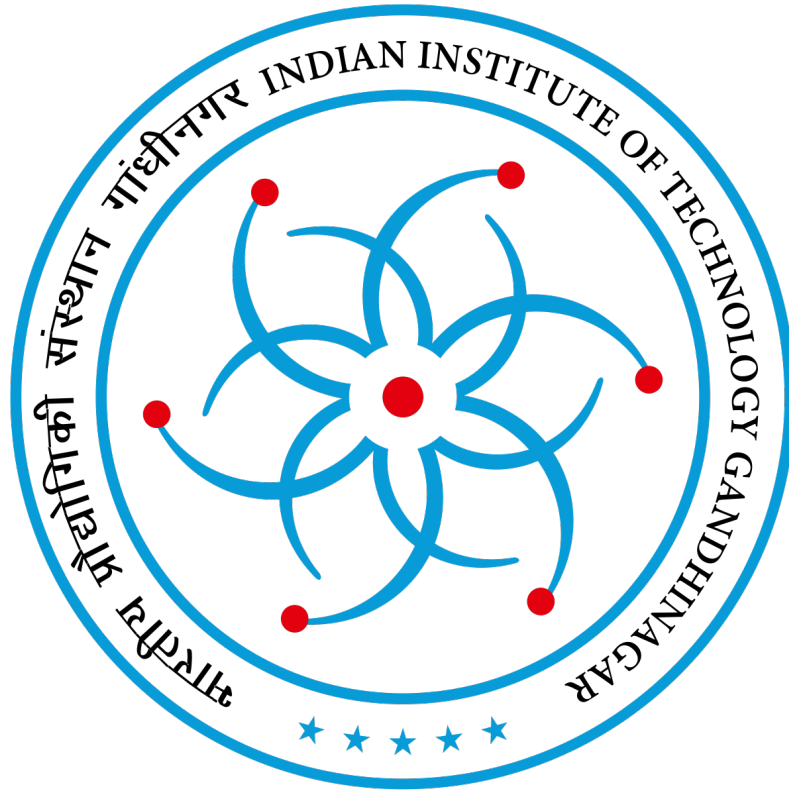


Indian Institute of Technology Gandhinagar



Computer Solution of 2-D Unsteady Heat Diffusion Equation

ME:605 Computational Fluid Dynamics

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1 Problem Statement

The aim of this project is to get the numerical solution of a Parabolic PDE, which in our case is the **Two Dimensional Unsteady Heat diffusion equation**, using the finite difference method, and get the solution using different approaches and Algorithms, and comment on the efficiency of one method over the other.

The governing differential equation for the problem statement is:-

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

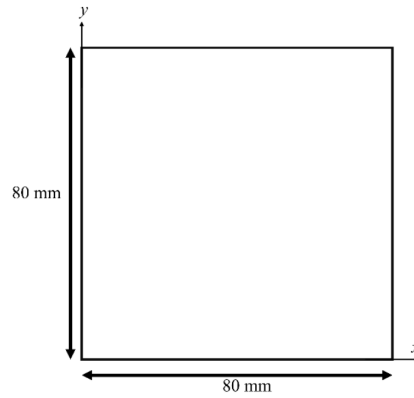


Figure 1: Square computational domain of size 80 mm $\times$ 80 mm.

1.1 Boundary Conditions

The boundary conditions are as follows:-

- $T(x, y = 0) = 323 \text{ K}$
- $T(x, y = 80 \text{ mm}) = 423 \text{ K}$
- $T(x = 80 \text{ mm}, y) = 473 \text{ K}$
- $h(T_\infty - T(x = 0, y)) = -\lambda \left. \frac{\partial T}{\partial x} \right|_{x=0}$
- $h = 250 \text{ W/m}^2\text{K}$, $\lambda = 5 \text{ W/m K}$, $T_\infty = 573\text{K}$ & $\alpha = 5 * 10^{-6} \text{ m}^2/\text{s}$
- The initial temperature of the domain is 300K .

2 Mesh Details

2.1 Mesh Type and Grid points

The type of grid used for the two-dimensional heat diffusion equation is a **Structured Grid**.

2.2 Grid Points

The number of Grid points in this computation is 81.

2.3 Domain Size

The domain size is the dimensions of the domain in which we are interested in calculating the temperatures. For this project, the domain size is 80 mm x 80 mm.

2.4 Grid Spacing

The grid spacing is the distance between two adjacent grid points. In this case, we have a structured and symmetric grid, meaning the grid spacing in the x and y directions is equal in magnitude.

The formula for calculating the grid spacing is as follows:-

- Firstly, discretize the mesh into smaller grids of size Δx and Δy .

$$\Delta x = \frac{\text{Domain Size}}{N - 1}, \Delta y = \frac{\text{Domain Size}}{N - 1}$$

(here N is the number of grid points and the size of the domain is the length of the domain).

3 Methodology for Discretization

3.1 Discretisation of Governing Equation for the explicit scheme

The governing Partial Differential equation for the 2-D unsteady heat diffusion equation is shown below.

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

Using the **Finite Difference Method** and discretising the time derivative term using the **Forward Difference Method**, we can convert the partial differential equations into algebraic equations. This type of discretisation is called the explicit method.

The forward difference discretisation for the time derivative term in the equation is shown below.

$$\left. \frac{dT}{dt} \right|^n = \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t}$$

The central difference discretisation for the space derivative term in the equation is shown below.

$$\begin{aligned} \left. \frac{d^2T}{dx^2} \right|^n &= \frac{T_{i+1,j}^n - 2T_{i,j}^n + T_{i-1,j}^n}{\Delta x^2} + O(\Delta x^2) \\ \left. \frac{d^2T}{dy^2} \right|^n &= \frac{T_{i,j+1}^n - 2T_{i,j}^n + T_{i,j-1}^n}{\Delta y^2} + O(\Delta y^2) \end{aligned}$$

We add these equations together, ignoring the error term, and since $\Delta x = \Delta y$, we get the following expression.

$$\begin{aligned} \left. \frac{d^2T}{dx^2} \right|^n + \left. \frac{d^2T}{dy^2} \right|^n &= \frac{T_{i+1,j}^n - 2T_{i,j}^n + T_{i-1,j}^n}{\Delta x^2} + \frac{T_{i,j+1}^n - 2T_{i,j}^n + T_{i,j-1}^n}{\Delta y^2} \\ \left. \frac{d^2T}{dx^2} \right|^n + \left. \frac{d^2T}{dy^2} \right|^n &= \frac{T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n - 4T_{i,j}^n}{\Delta x^2} \end{aligned}$$

Assembling the equations, we get the following expression.

$$\begin{aligned} \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} &= \alpha \frac{(T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n - 4T_{i,j}^n)}{\Delta x^2} \\ T_{i,j}^{n+1} &= T_{i,j}^n + \frac{\alpha \Delta t}{\Delta x^2} (T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n - 4T_{i,j}^n) \end{aligned}$$

The final discretised equations for the explicit scheme are shown below.

$$\boxed{\mathbf{T}_{i,j}^{n+1} = \mathbf{T}_{i,j}^n + \frac{\alpha \Delta t}{\Delta x^2} (\mathbf{T}_{i+1,j}^n + \mathbf{T}_{i-1,j}^n + \mathbf{T}_{i,j+1}^n + \mathbf{T}_{i,j-1}^n - 4\mathbf{T}_{i,j}^n)}$$

The variables represent the following physical quantity

- α is the thermal diffusivity.
- Δt is the time step.
- Δx is the grid spacing.
- $n + 1$ is the current time step.
- n is the previous time step.

3.2 Discretisation of Governing Equation for the implicit scheme

The governing Partial Differential equation for the 2-D unsteady heat diffusion equation is shown below.

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

Using the **Finite Difference Method** and discretising the time derivative term using the **Backward Difference Method**, we can convert the partial differential equations into algebraic equations. This type of discretisation is called the implicit method.

The backward difference discretisation for the time derivative term in the equation is shown below.

$$\left. \frac{dT}{dt} \right|^{n+1} = \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t}$$

The central difference discretisation for the space derivative term in the equation is shown below.

$$\begin{aligned} \left. \frac{d^2 T}{dx^2} \right|^{n+1} &= \frac{T_{i+1,j}^{n+1} - 2T_{i,j}^{n+1} + T_{i-1,j}^{n+1}}{\Delta x^2} + O(\Delta x^2) \\ \left. \frac{d^2 T}{dy^2} \right|^{n+1} &= \frac{T_{i,j+1}^{n+1} - 2T_{i,j}^{n+1} + T_{i,j-1}^{n+1}}{\Delta y^2} + O(\Delta y^2) \end{aligned}$$

We add these equations together, ignoring the error term, and since $\Delta x = \Delta y$, we get the following expression.

$$\begin{aligned} \left. \frac{d^2 T}{dx^2} \right|^{n+1} + \left. \frac{d^2 T}{dy^2} \right|^{n+1} &= \frac{T_{i+1,j}^{n+1} - 2T_{i,j}^{n+1} + T_{i-1,j}^{n+1}}{\Delta x^2} + \frac{T_{i,j+1}^{n+1} - 2T_{i,j}^{n+1} + T_{i,j-1}^{n+1}}{\Delta y^2} \\ \left. \frac{d^2 T}{dx^2} \right|^{n+1} + \left. \frac{d^2 T}{dy^2} \right|^{n+1} &= \frac{T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} - 4T_{i,j}^{n+1}}{\Delta x^2} \end{aligned}$$

Assembling the equations, we get the following expression.

$$\begin{aligned} \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} &= \alpha \frac{(T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} - 4T_{i,j}^{n+1})}{\Delta x^2} \\ T_{i,j}^{n+1} &= T_{i,j}^n + \frac{\alpha \Delta t}{\Delta x^2} (T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} - 4T_{i,j}^{n+1}) \end{aligned}$$

Let $\frac{\alpha \Delta t}{\Delta x^2}$ be replaced by F_o (*Fourier's Number*).

$$T_{i,j}^{n+1} = T_{i,j}^n + F_o (T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} - 4T_{i,j}^{n+1})$$

The final form of the discretised equation is as follows:-

$$(1 + 4F_o)T_{i,j}^{n+1} - F_o(T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} + T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1}) = T_{i,j}^n$$

The variables represent the following physical quantity

- F_o is the Fourier's Number.
- $n + 1$ is the current time step.
- n is the previous time step.

3.3 Discretisation of Governing Equation for the Crank-Nicolson scheme

The governing Partial Differential equation for the 2-D unsteady heat diffusion equation is shown below.

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

Using the **Finite Difference Method** and discretising the time derivative and spatial derivative in a way to get second order accurate in space and time terms, such a type of discretisation leads us to the Crank-Nicolson method.

The central difference discretisation for the time derivative term in the equation is shown below.

$$\left. \frac{dT}{dt} \right|^{n+\frac{1}{2}} = \frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t}$$

The central difference discretisation for the space derivative term in the equation is shown below.

$$\begin{aligned} \left. \frac{d^2 T}{dx^2} \right|^{n+\frac{1}{2}} &= \frac{T_{i+1,j}^{n+\frac{1}{2}} - 2T_{i,j}^{n+\frac{1}{2}} + T_{i-1,j}^{n+\frac{1}{2}}}{\Delta x^2} + O(\Delta x^2) \\ \left. \frac{d^2 T}{dy^2} \right|^{n+\frac{1}{2}} &= \frac{T_{i,j+1}^{n+\frac{1}{2}} - 2T_{i,j}^{n+\frac{1}{2}} + T_{i,j-1}^{n+\frac{1}{2}}}{\Delta y^2} + O(\Delta y^2) \\ T^{n+\frac{1}{2}} &= \frac{T^n + T^{n+1}}{2} \end{aligned}$$

We substitute the above relation in the equation to get the following formula. Add these equations together, ignoring the error term. Since $\Delta x = \Delta y$, we get the following expression.

$$\frac{T_{i,j}^{n+1} - T_{i,j}^n}{\Delta t} = \frac{\alpha}{2(\Delta x)^2} \left[\left(T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} - 4T_{i,j}^{n+1} \right) \right. \\ \left. + \left(T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n - 4T_{i,j}^n \right) \right]$$

$$T_{i,j}^{n+1} - T_{i,j}^n = \frac{\alpha \Delta t}{2(\Delta x)^2} \left[\left(T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} - 4T_{i,j}^{n+1} \right) \right. \\ \left. + \left(T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n - 4T_{i,j}^n \right) \right]$$

We substitute the expression $\frac{\alpha \Delta t}{2\Delta x^2}$ with $\mathbf{F}'_o$ (Modified Fourier's Number).

$$T_{i,j}^{n+1} - T_{i,j}^n = F'_o \left[\left(T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} - 4T_{i,j}^{n+1} \right) \right. \\ \left. + \left(T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n - 4T_{i,j}^n \right) \right]$$

The final discretised equation for the 2D unsteady heat conduction equation using the Crank-Nicolson scheme is:

$$\boxed{\begin{aligned} (1 + 4F'_o) T_{i,j}^{n+1} - F'_o (T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1}) = \\ (1 - 4F'_o) T_{i,j}^n + F'_o (T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n) \end{aligned}}$$

The variables represent the following physical quantity

- $\mathbf{F}'_o$ is the Modified Fourier's Number ($\mathbf{F}'_o = \mathbf{F}_o/2$).
- $\mathbf{n} + 1$ is the current time step.
- $\mathbf{n}$ is the previous time step.

3.4 Dirichlet Boundary Conditions

The Boundary Conditions specified at the domain's top, bottom and right boundary are Dirichlet Boundary Conditions, meaning the temperature at these boundaries does not change.

3.5 Neumann boundary condition

The Boundary Conditions specified at the domain's left boundary are Neumann Boundary Conditions, meaning the temperature at the left boundary will change accordingly.

The governing differential equation at the left boundary is as follows:-

$$h(T_{\infty} - T(x=0, y)) = -\lambda \left. \frac{\partial T}{\partial x} \right|_{x=0}$$

We use a first-order approximation to convert the following differential equations into an algebraic equation.

As we are at the left boundary, we have no point to the left, but we have a point to the right of the boundary, so instead of using the Central difference scheme, we use the forward difference scheme to discretise the equation.

According to the first-order approximation of $\frac{\partial T}{\partial x}$ we get:

$$\frac{\partial T}{\partial x} = \frac{T_{i,j+1} - T_{i,j}}{\Delta x}$$

$$h(T_{\infty} - T(x=0, y)) = -\lambda \frac{T_{i,2} - T_{i,1}}{\Delta x}$$

$$h(T_{\infty} - T_{i,1}) = -\lambda \frac{T_{i,2} - T_{i,1}}{\Delta x}$$

After simplifying the expression, we get the following expression.

$$T_{i,1} = \frac{h\Delta x T_{\infty} + \lambda T_{i,2}}{\lambda + h\Delta x}$$

The variables represent the following physical quantity

- Only for the Left boundary i, j represent the row and column respectively.
- h is the convection heat transfer coefficient.
- λ is the thermal conductivity.
- T_{∞} is the ambient temperature.
- Δx is the grid spacing.
- $T_{i,1}$ is the unknown temperature of the left boundary.
- $T_{i,2}$ is the temperature of the grid point to the right of $T_{i,1}$.

4 Solution Methodology

The Gauss–Seidel method is an iterative method used to solve a system of linear equations. Though it can be applied to any matrix with non-zero elements on the diagonals, convergence is only guaranteed if the matrix is either strictly diagonally dominant or symmetric and positive definite.

A general 3 by 3 matrix looks like this.

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \quad X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad B = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

When $\mathbf{A}$ and $\mathbf{b}$ are known, and $\mathbf{x}$ is unknown, the Gauss-Seidel method can be used to iteratively approximate $\mathbf{x}$. The vector $\mathbf{x}^{(0)}$ denotes the initial guess for $\mathbf{x}$, often $\mathbf{x}_i^{(0)} = 0$ for $i = 1, 2, \dots, n$. Denote by $\mathbf{x}^{(k)}$ the k -th approximation (or iteration) of $\mathbf{x}$, and by $\mathbf{x}^{(k+1)}$ the approximation of $\mathbf{x}$ at the next or $k + 1$ -th iteration.

In Gauss-Seidel, the computation of $\mathbf{x}^{(k+1)}$ uses the elements of $\mathbf{x}^{(k+1)}$ that have already been computed, and only the elements of $\mathbf{x}^{(k)}$ that have not yet been computed in the $(k + 1)$ -th iteration. This means that, unlike the Jacobi method, only one storage vector is required since elements can be overwritten as they are computed, which can be advantageous for very large problems.

The convergence properties of the Gauss-Seidel method are dependent on the matrix $\mathbf{A}$. Namely, the procedure is known to converge if either:

- $\mathbf{A}$ is symmetric positive-definite, or
- $\mathbf{A}$ is strictly or irreducibly diagonally dominant.

The Gauss-Seidel method may converge even if these conditions are not satisfied.

4.1 Explicit Scheme

4.1.1 Solution Methodology for the Explicit Scheme

The discretised governing equation is shown below.

$$\mathbf{T}_{ij}^{n+1} = \mathbf{T}_{ij}^n + \frac{\alpha \Delta t}{\Delta x^2} (\mathbf{T}_{i+1,j}^n + \mathbf{T}_{i-1,j}^n + \mathbf{T}_{i,j+1}^n + \mathbf{T}_{i,j-1}^n - 4\mathbf{T}_{ij}^n)$$

When we examine the discretised equations at each time step, we find that we can determine the temperature of each node independently. This means we do not have to iteratively calculate the temperature for a time step for any of the nodes. However, as we progress to a different time step, the temperature of the domain will continue to change until we reach a steady state. To verify whether our solution has reached a steady state, we check the temperature difference between two time steps and compare its $L2$ norm with the tolerance set by us.

We first create a zero matrix of the size N where N is the number of grid points and assign the known temperatures to the top, right and bottom boundary and the initial temperature to the remaining nodes. The temperature matrix for five grid points will look like the matrix depicted below.

$$\mathbf{T} = \begin{bmatrix} 423 & 423 & 423 & 423 & 473 \\ 300 & 300 & 300 & 300 & 473 \\ 300 & 300 & 300 & 300 & 473 \\ 300 & 300 & 300 & 300 & 473 \\ 323 & 323 & 323 & 323 & 473 \end{bmatrix}$$

For the interior nodes, the temperature of the nodes at every time step is calculated according to the following formula.

$$\mathbf{T}_{ij}^{n+1} = \mathbf{T}_{ij}^n + \frac{\alpha \Delta t}{\Delta x^2} (\mathbf{T}_{i+1,j}^n + \mathbf{T}_{i-1,j}^n + \mathbf{T}_{i,j+1}^n + \mathbf{T}_{i,j-1}^n - 4\mathbf{T}_{ij}^n)$$

- n is the previous time-step.
- $n + 1$ is the current time-step.

For the nodes on the left boundary, the temperature is calculated using a first-order approximation of the Neumann boundary condition, and the discretised equation for the same is shown below.

$$T_{i,1}^{n+1} = \frac{h\Delta x T_{\infty} + \lambda T_{i,2}^n}{\lambda + h\Delta x}$$

- n is the previous time-step.
- $n + 1$ is the current time-step.

4.1.2 Working Principle of the Code

The workflow of the code has been explained below:-

1. Define the necessary variables and parameters required.
2. Create the initial temperature matrix with the given temperatures at boundaries and nodes.
3. The explicit scheme calculates the temperature of every node accurately for a time step and stores it.
4. We calculate the difference in temperature between the newly calculated temperature matrix and the previously calculated temperature matrix, take the $L2$ Norm and compare it with the tolerance level set.
5. This process is looped over, the temperature is calculated after every time step, and the loop terminates when the temperature difference between two consecutive time steps is less than the tolerance.

4.1.3 Steady State Temperature Field for N = 81 grid points

The temperature distribution using the explicit scheme method for 81 grid points in x and y directions has been shown below.

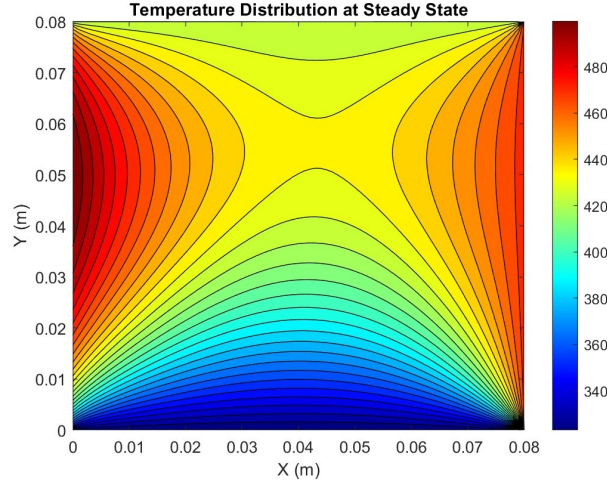


Figure 2: Temperature distribution for N = 81 grid points using the explicit scheme

4.1.4 Stability Check of the explicit scheme

As derived using *von Neumann stability analysis*, the FTCS (forward time-centred space) method for the one-dimensional heat equation is numerically stable if and only if the following condition is satisfied:

$$\Delta t \leq \frac{\Delta x^2}{2\alpha}.$$

Which is to say that the choice of Δx and Δt must satisfy the above condition for the FTCS scheme to be stable. In two dimensions, the condition becomes

$$\Delta t \leq \frac{1}{2\alpha \left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} \right)}.$$

If we choose $\delta = \Delta x = \Delta y$ then the stability conditions for two-dimensional applications becomes

$$\Delta t \leq \frac{\delta^2}{4\alpha}$$

A significant drawback of the FTCS(forward time-centred space) method is that for problems with large diffusivity α , satisfactory step sizes can be too small to be practical.

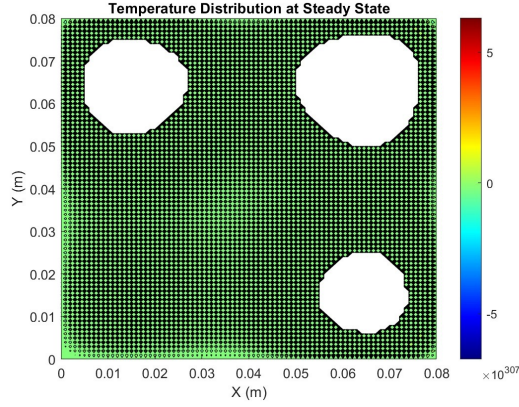


Figure 3: Temperature distribution for $N = 81$ grid points using the explicit scheme

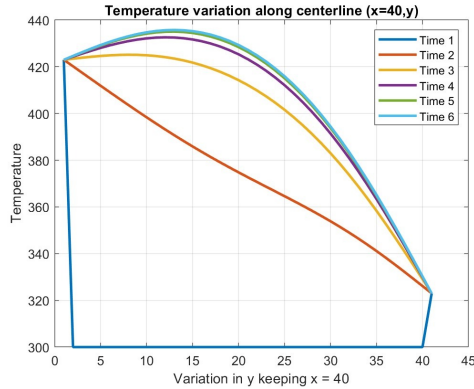
The temperature distribution shown above is obtained when the stability criteria condition is violated, which results in an incorrect temperature distribution, and in the above case, temperatures at nodes are not even defined, and the net distribution is wrong.

4.1.5 Critical Time step beyond which methods become unstable

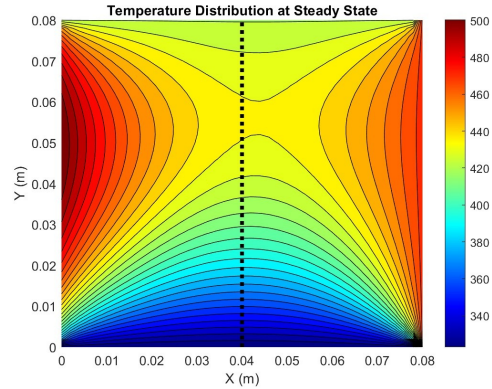
For 2 – D domain the formula for calculating $\Delta t_{critical}$ is given above. The value of the parameter in the equation for our problem is $\alpha = 5 * 10^{-6} \text{ m}^2/\text{s}$ and $h = 250\text{W}/\text{m}^2\text{k}$. Substituting the values of the variables in the equation above, we get the critical time step.

$$\Delta t_{critical} = \frac{\delta^2}{4\alpha} = \frac{(0.08/80)^2}{4(5 \times 10^{-6})} = 0.05 \text{ s}$$

4.1.6 Temperature Variation along centre lines ($x = 40\text{mm}, y$)



(a) Temperature along $x = 40\text{mm}, y$



(b) Steady State Temperature distribution

Figure 4: Temperature variation along the centre lines ($x = 40\text{mm}, y$) of the domain.

The graph plotted above shows the distribution of temperature along the centre line ($x = 40\text{mm}, y$) for six equally spaced time intervals. Analysing the graph, we can first see that the variation in temperature for interior nodes decreases as we approach the steady state. In contrast, points on the Top & Bottom boundaries are unaffected as we have given Dirichlet conditions. Secondly, the temperature near the midpoint of the centre line reaches a maximum. Then it starts decreasing again, and this can be explained by drawing a vertical line from the centre of the steady-state contour.

4.1.7 Temperature Variation along centre lines ($x, y = 40\text{mm}$)

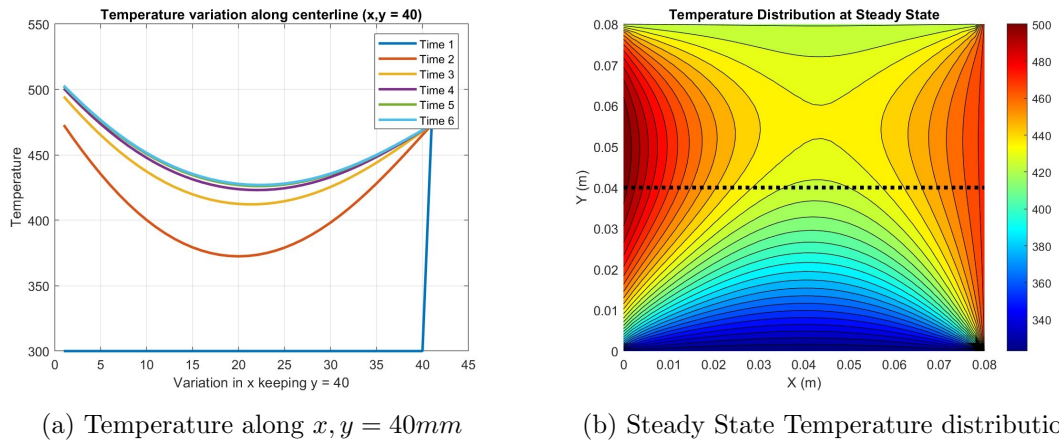


Figure 5: Temperature variation along the centre lines ($x, y = 40\text{mm}$) of the domain.

The graph plotted above shows the distribution of temperature along the centre line ($x, y = 40\text{mm}$) for six equally spaced time intervals. Analysing the graph, we can first see that the variation in temperature for interior nodes and nodes on the Left Boundary (*Neumann Boundary Condition*) decreases as we approach the steady state. In contrast, points on the are unaffected as we have given Dirichlet conditions. Secondly, the temperature near the midpoint of the centre line reaches a minimum and then starts increasing again, and this can be explained by drawing a vertical line from the centre of the steady-state contour.

4.2 Implicit Scheme

4.2.1 Solution Methodology for Implicit Scheme

The discretised governing equation is shown below.

$$(1 + 4F_o)T_{i,j}^{n+1} - F_o(T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1} + T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1}) = T_{i,j}^n$$

When we examine the discretised equations at each time step, we find that to calculate the temperature value of a node in the current time step, we also simultaneously need to

calculate the temperature at the neighbouring nodes in the current time step, meaning these equations are coupled. To solve them, we have multiple approaches, one of which includes forming the matrix A and vector B and solving the linear equations for every time step until the temperature variation is less than the tolerance set by us, which will be computationally expensive. The second approach includes using two loops, one which calculates the error while iterating within a time step, and the second that calculates the error between temperature matrices obtained between two time steps. This process will be very fast as we don't need to form and solve the huge matrices. The implicit scheme takes into consideration the actual physics of the problem, whereas the explicit scheme does not. This unique property of the scheme makes it unconditionally stable and appropriate to solve problems.

We first create a zero matrix of the size N where N is the number of grid points and assign the known temperatures to the top, right and bottom boundary and the initial temperature to the remaining nodes. The temperature matrix for five grid points will look like the matrix depicted below.

$$\mathbf{T} = \begin{bmatrix} 423 & 423 & 423 & 423 & 473 \\ 300 & 300 & 300 & 300 & 473 \\ 300 & 300 & 300 & 300 & 473 \\ 300 & 300 & 300 & 300 & 473 \\ 323 & 323 & 323 & 323 & 473 \end{bmatrix}$$

For the interior nodes, the temperature of the nodes at every iteration is calculated according to the following formula.

$$\mathbf{T}_{ij}^{n+1} = \frac{\mathbf{T}_{ij}^n + \mathbf{F}_o(\mathbf{T}_{i,j+1}^{n+1} + \mathbf{T}_{i,j-1}^{n+1} + \mathbf{T}_{i+1,j}^{n+1} + \mathbf{T}_{i-1,j}^{n+1})}{(1 + 4\mathbf{F}_o)}$$

The modified equation for the interior nodes using the implicit scheme is shown below.

$$\mathbf{T}_{ij}^{k+1} = \frac{\mathbf{T}_{ij}^n + \mathbf{F}_o(\mathbf{T}_{i,j+1}^{k+1} + \mathbf{T}_{i,j-1}^{k+1} + \mathbf{T}_{i+1,j}^{k+1} + \mathbf{T}_{i-1,j}^{k+1})}{(1 + 4\mathbf{F}_o)}$$

- n is the previous time-step.
- $k + 1$ denotes the current Gauss–Seidel iteration. The value of the temperature at the node at this stage is still approximate. As iterations proceed and the residual decreases, the solution approaches $n + 1$. When the residual is sufficiently small, the $k + 1$ iteration is treated as the final iteration $n + 1$.
- $n + 1$ is the current time-step.

For the nodes on the left boundary, the temperature is calculated using a first-order approximation of the Neumann boundary condition, and the discretised equation for the same is shown below.

$$T_{i,1}^{n+1} = \frac{h\Delta x T_{\infty} + \lambda T_{i,2}^{n+1}}{\lambda + h\Delta x}$$

The modified equation for the nodes on the Left boundary using the implicit scheme is shown below.

$$T_{i,1}^{k+1} = \frac{h\Delta x T_{\infty} + \lambda T_{i,2}^{k+1}}{\lambda + h\Delta x}$$

- n is the previous time-step.
- $k + 1$ denotes the current Gauss–Seidel iteration. The value of the temperature at the node at this stage is still approximate. As iterations proceed and the residual decreases, the solution approaches $n + 1$. When the residual is sufficiently small, the $k + 1$ iteration is treated as the final iteration $n + 1$.
- $n + 1$ is the current time-step.

4.2.2 Working Principle of the code

The workflow of the code has been explained below:-

1. Define the necessary variables and parameters required.
2. Initialise the temperature matrix T with the given boundary temperatures and the initial temperature for the interior nodes.
3. Calculate the temperatures for interior nodes and nodes of the left boundary for $(k + 1)$ iteration using the modified equations.
4. Calculate the residual by taking the $L2$ norm of the difference of the temperature matrix obtained at $(k + 1)^{th}$ iteration and n^{th} time-step.
5. If the residual is less than the tolerance set by us, then $(k + 1)^{th}$ iteration equals the temperature of $(n + 1)^{th}$ time-step.
6. Looping this process over, we can get the temperature at the steady state when the error calculated by taking the $L2$ norm of the difference of the temperature matrix obtained at $(n + 1)^{th}$ time step and n^{th} time-step is less than the error set by us.

4.2.3 Steady State Temperature Field for $N = 81$ grid points

The temperature distribution using the implicit scheme method for 81 grid points in x and y directions has been shown below.

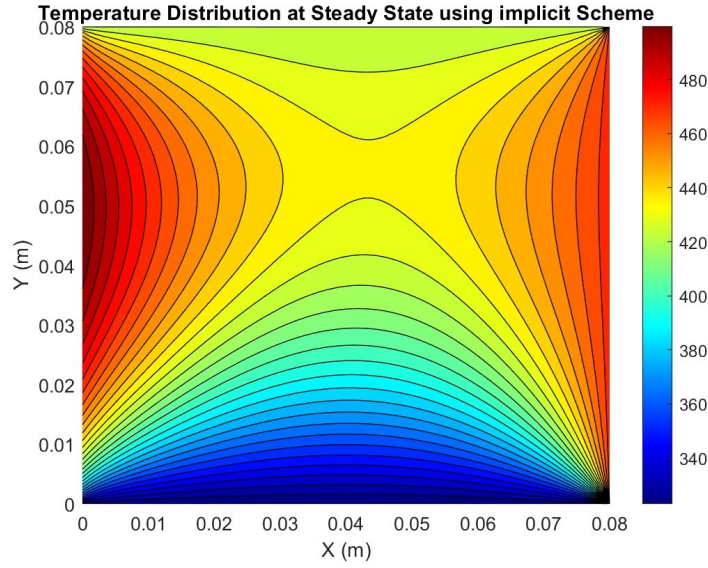
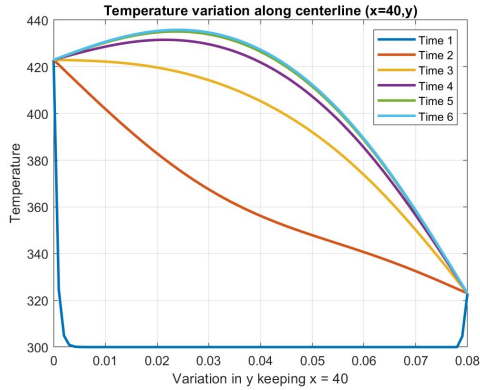
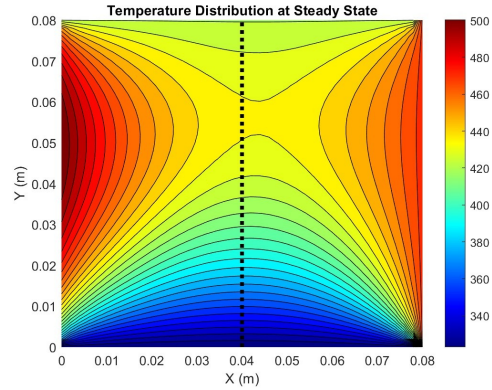


Figure 6: Temperature distribution for $N = 81$ grid points using the implicit scheme

4.2.4 Temperature Variation along centre lines ($x = 40\text{mm}, y$)



(a) Temperature along $x = 40\text{mm}, y$



(b) Steady State Temperature distribution

Figure 7: Temperature variation along the centre lines ($x = 40\text{mm}, y$) of the domain.

The graph plotted above shows the distribution of temperature along the centre line ($x = 40\text{mm}, y$) for six equally spaced time intervals. Analysing the graph, we can first see that the variation in temperature for interior nodes decreases as we approach the steady state. In contrast, points on the Top & Bottom boundaries are unaffected as we have given Dirichlet conditions. Secondly, the temperature near the midpoint of the centre line reaches a maximum and then starts decreasing again, and this can be explained by drawing a vertical line from the centre of the steady-state contour.

4.2.5 Temperature Variation along centre lines ($x, y = 40\text{mm}$)

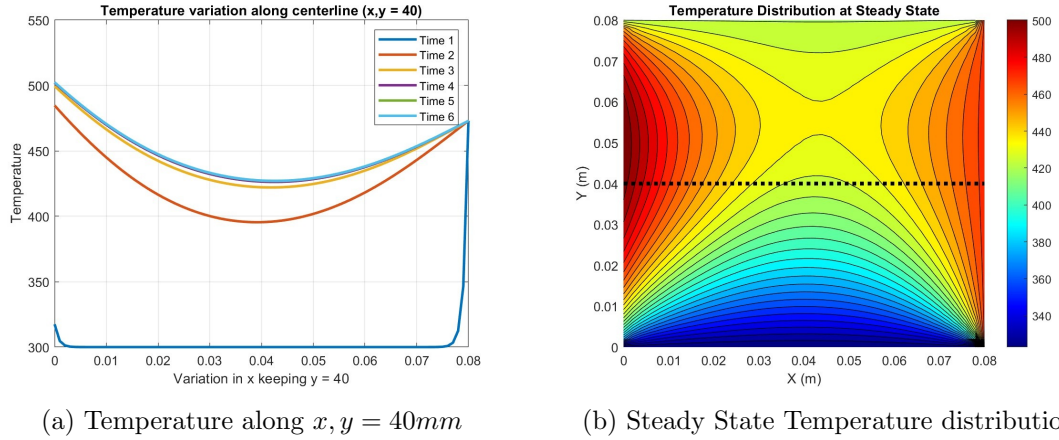


Figure 8: Temperature variation along the centre lines ($x, y = 40\text{mm}$) of the domain.

The graph plotted above shows the distribution of temperature along the centre line ($x, y = 40\text{mm}$) for six equally spaced time intervals. Analysing the graph, we can first see that the variation in temperature for interior nodes and nodes on the Left Boundary (*Neumann Boundary Condition*) decreases as we approach the steady state. In contrast, points on the right boundary are unaffected as we have given Dirichlet conditions. Secondly, the temperature near the midpoint of the centre line reaches a minimum and then starts increasing again, and this can be explained by drawing a vertical line from the centre of the steady-state contour.

4.3 Crank-Nicolson Scheme

4.3.1 Solution Methodology for Crank-Nicolson Scheme

The discretised governing equation is shown below.

$$(1 + 4F_o') T_{i,j}^{n+1} - F_o' (T_{i+1,j}^{n+1} + T_{i-1,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^{n+1}) = (1 - 4F_o') T_{i,j}^n + F_o' (T_{i+1,j}^n + T_{i-1,j}^n + T_{i,j+1}^n + T_{i,j-1}^n)$$

When we examine the discretised equations at each time step, we find that to calculate the temperature value of a node in the current time step, we also simultaneously need to calculate the temperature at the neighbouring nodes in the current time step, meaning these equations are coupled. To solve them, we have multiple approaches, one of which includes forming the matrix A and vector B and solving the linear equations for every time step until the temperature variation is less than the tolerance set by us, which will be computationally expensive. The second approach includes using two loops, one which calculates the error while iterating within a time step, and the second that calculates the

error between temperature matrices obtained between two time steps. This process will be very fast as we don't need to form and solve the huge matrices. The Crank-Nicolson scheme takes into consideration the actual physics of the problem, whereas the explicit scheme does not. This unique property of the scheme makes it unconditionally stable and appropriate to solve problems.

$$\mathbf{T} = \begin{bmatrix} 423 & 423 & 423 & 423 & 473 \\ 300 & 300 & 300 & 300 & 473 \\ 300 & 300 & 300 & 300 & 473 \\ 300 & 300 & 300 & 300 & 473 \\ 323 & 323 & 323 & 323 & 473 \end{bmatrix}$$

For the interior nodes, the temperature of the nodes at every time step is calculated according to the following formula.

$$\begin{aligned} (1 + 4F'_o) \mathbf{T}_{i,j}^{n+1} &= (1 - 4F'_o) \mathbf{T}_{i,j}^n \\ &+ F'_o (\mathbf{T}_{i+1,j}^n + \mathbf{T}_{i-1,j}^n + \mathbf{T}_{i,j+1}^n + \mathbf{T}_{i,j-1}^n) \\ &+ F'_o (\mathbf{T}_{i+1,j}^{n+1} + \mathbf{T}_{i-1,j}^{n+1} + \mathbf{T}_{i,j+1}^{n+1} + \mathbf{T}_{i,j-1}^{n+1}) \end{aligned}$$

The modified equation for interior nodes using the Crank-Nicolson scheme is shown below.

$$\begin{aligned} (1 + 4F'_o) \mathbf{T}_{i,j}^{k+1} &= (1 - 4F'_o) \mathbf{T}_{i,j}^n \\ &+ F'_o (\mathbf{T}_{i+1,j}^n + \mathbf{T}_{i-1,j}^n + \mathbf{T}_{i,j+1}^n + \mathbf{T}_{i,j-1}^n) \\ &+ F'_o (\mathbf{T}_{i+1,j}^{k+1} + \mathbf{T}_{i-1,j}^{k+1} + \mathbf{T}_{i,j+1}^{k+1} + \mathbf{T}_{i,j-1}^{k+1}) \end{aligned}$$

- F'_o is the Modified Fourier's Number ($F'_o = F_o/2$)
- n is the previous time-step.
- $(k+1)$ denotes the current Gauss-Seidel iteration. The value of the temperature at the node at this stage is still approximate. As iterations proceed and the residual decreases, the solution approaches $n+1$. When the residual is sufficiently small, the $(k+1)$ iteration is treated as the final iteration ($n+1$).
- $(n+1)$ is the current time-step.

For the nodes on the left boundary, the temperature is calculated using a first-order approximation of the Neumann boundary condition, and the discretised equation for the same is shown below.

$$T_{i,1}^{n+1} = \frac{h\Delta x T_{\infty} + \lambda T_{i,2}^{n+1}}{\lambda + h\Delta x}$$

The modified equation for the nodes on the Left boundary using the Crank-Nicolson scheme is shown below.

$$T_{i,1}^{k+1} = \frac{h\Delta x T_{\infty} + \lambda T_{i,2}^{k+1}}{\lambda + h\Delta x}$$

- n is the previous time-step.
- $(k+1)$ denotes the current Gauss-Seidel iteration. The value of the temperature at the node at this stage is still approximate. As iterations proceed and the residual decreases, the solution approaches $(n+1)$. When the residual is sufficiently small, the $(k+1)$ iteration is treated as the final iteration $(n+1)$.
- $n+1$ is the current time-step.

4.3.2 Working Principle of the code

The workflow of the code has been explained below:-

1. Define the necessary variables and parameters required.
2. Initialise the temperature matrix T with the given boundary temperatures and the initial temperature for the interior nodes.
3. Calculate the temperatures for interior nodes and nodes of the left boundary for $(k+1)$ iteration using the modified equations.
4. Calculate the residual by taking the $L2$ norm of the difference of the temperature matrix obtained at $(k+1)^{th}$ iteration and n^{th} time-step.
5. If the residual is less than the tolerance set by us, then $(k+1)^{th}$ iteration equals the temperature of $(n+1)^{th}$ time-step.
6. Looping this process over, we can get the temperature at the steady state when the error calculated by taking the $L2$ norm of the difference of the temperature matrix obtained at $(n+1)^{th}$ time step and n^{th} time step is less than the error set by us.

4.3.3 Steady State Temperature Field for $N = 81$ grid points

The temperature distribution using the Crank-Nicolson scheme method for 81 grid points in x and y directions has been shown below.

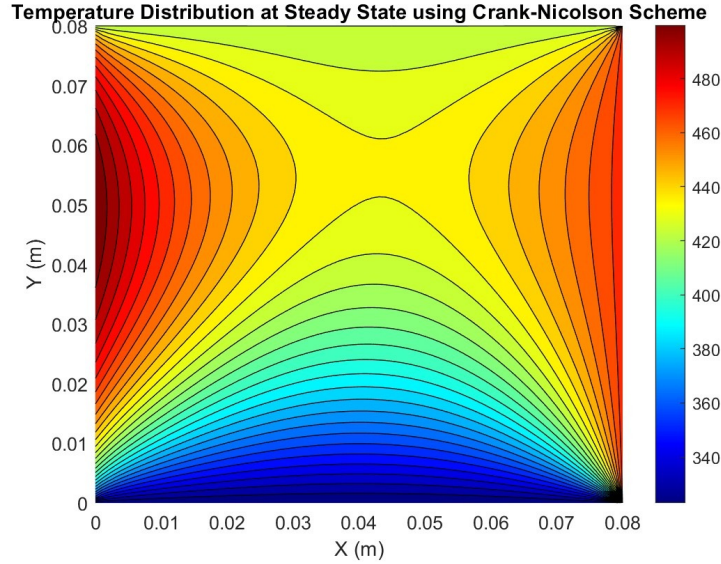


Figure 9: Temperature distribution for $N = 81$ grid points using the Crank-Nicolson scheme

4.3.4 Temperature Variation along centre lines ($x = 40\text{mm}, y$)

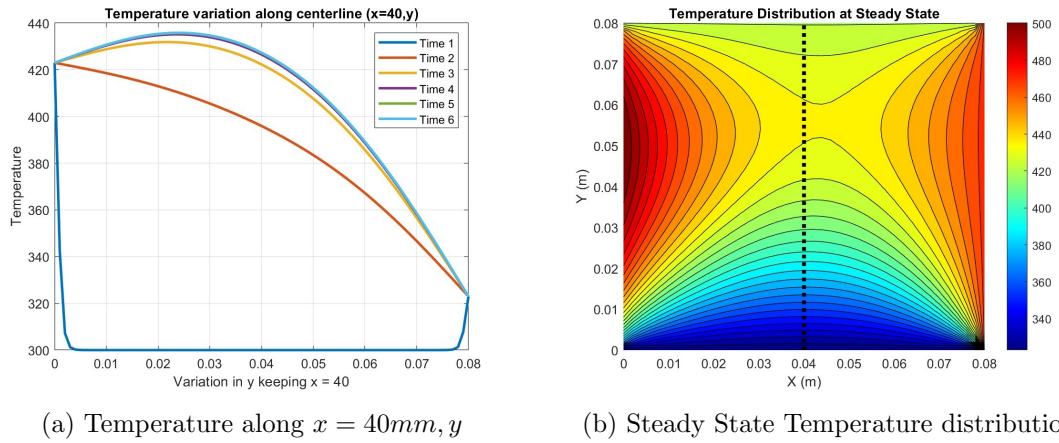


Figure 10: Temperature variation along the centre lines ($x = 40\text{mm}, y$) of the domain.

The graph plotted above shows the distribution of temperature along the centre line ($x = 40\text{mm}, y$) for six equally spaced time intervals. Analysing the graph, we can first

see that the variation in temperature for interior nodes decreases as we approach the steady state. In contrast, points on the Top & Bottom boundaries are unaffected as we have given Dirichlet conditions. Secondly, the temperature near the midpoint of the centre line reaches a maximum and then starts decreasing again, and this can be explained by drawing a vertical line from the centre of the steady-state contour. Thirdly, as we approach steady state, the temperature centre lines almost overlap, meaning the temperature variation is not significant.

4.3.5 Temperature Variation along centre lines ($x, y = 40\text{mm}$)

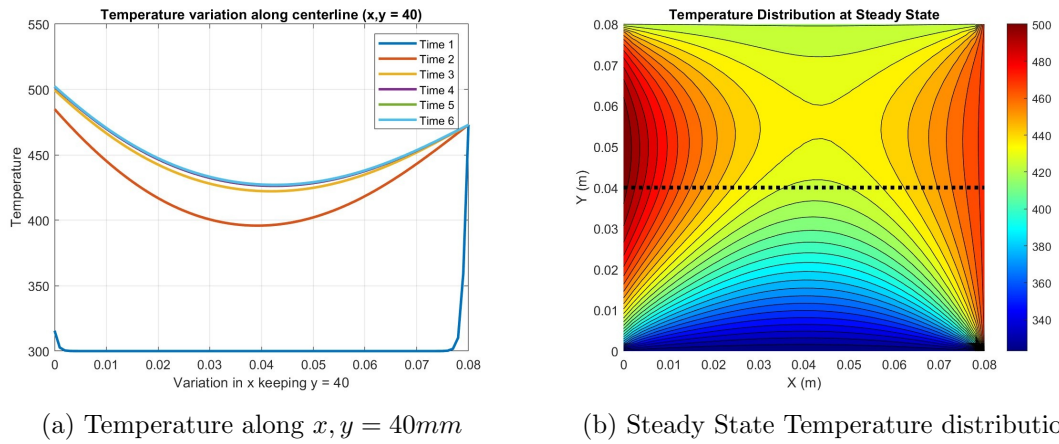


Figure 11: Temperature variation along the centre lines ($x, y = 40\text{mm}$) of the domain.

The graph plotted above shows the distribution of temperature along the centre line ($x, y = 40\text{mm}$) for six equally spaced time intervals. Analysing the graph, we can first see that the variation in temperature for interior nodes and nodes on the Left Boundary (*Neumann Boundary Condition*) decreases as we approach the steady state. In contrast, points on the right boundary are unaffected as we have given Dirichlet conditions. Secondly, the temperature near the midpoint of the centre line reaches a minimum and then starts increasing again, and this can be explained by drawing a vertical line from the centre of the steady-state contour.

5 Results and Discussions

1. The final steady-state temperature contours obtained using the Explicit, Implicit, and Crank–Nicolson methods are nearly identical.
2. The Explicit scheme was the fastest compared to the Implicit and Crank-Nicolson methods, as the discretised equation was not coupled and the temperature at any node could be calculated independently, and it involved a lesser number of calculations.
3. The Explicit scheme converged only when the stability criteria were satisfied, whereas the Implicit and Crank-Nicolson methods were unconditionally stable.
4. The Explicit scheme does not take into consideration the physics of the problem, whereas the Implicit and Crank-Nicolson methods took care of the physics of the problem, as the temperature of any node depended on the temperature of all its neighbours.
5. The Crank-Nicolson is second-order accurate in time and space, whereas the Implicit scheme is second-order accurate in space but first-order accurate in time. Hence, the Crank-Nicolson will give us a better sense of the diffusion of temperature over time than the Implicit scheme.

6 Concluding Remarks

This project helped me understand the numerical solution of parabolic PDEs and provided insight into how the domain of dependence of the scheme affects its stability. One important thing I learned is that the Implicit and Crank-Nicolson schemes were slower than the explicit scheme but they were also unconditionally stable. Overall, the project highlighted the importance of striking a balance between the accuracy and computational efficiency of a numerical algorithm.

7 Acknowledgments and References

I would like to express my gratitude to Professor Dilip Sundaram for teaching us the concepts of Computational Fluid Dynamics clearly and concisely, and for assigning this project to practically test our understanding of the concepts we were taught. Along with this, I would like to thank our Teaching Assistants, Mr Pardha Sai Mutyala and Mr Soham Pathak, for teaching us different algorithms during the tutorial sessions and helping me clarify any questions that arose while completing this project.

The following sources were referred to.

1. https://en.wikipedia.org/wiki/FTCS_scheme
2. https://en.wikipedia.org/wiki/Gauss%E2%80%93Seidel_method